

Delivery of Engineering Processes Safely As A Team. Manufacture of a Batch of Five Aluminium Spirit Levels.

Unit 2 - Delivery of Engineering Processes Safely as a Team.

Manufacture of a Batch of Five Aluminium Spirit Levels.

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Project: Manufacture of five aluminium spirit levels.

Batch size: Five units.

Material: Aluminium Ø30 mm × 60 mm.

Team Size: 5 Members.

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1. Introduction.

Purpose of the Project.

To be honest, when I first saw the engineering drawing for the spirit level, I thought it looked pretty straightforward. A bit of turning, some milling, drill a few holes, tap some threads, how hard could it be? But once we actually got into the workshop and picked up the aluminium stock, I quickly realised there was more to it than I had expected.

This project was about manufacturing a batch of five aluminium spirit levels, working as part of a five-person team. We had to use a combination of lathe operations, milling, and bench fitting. What made it interesting was the leadership rotation, every one of us had to take a turn leading the team. That part actually made me more nervous than the machining itself, if I am honest. I had never led a team in a workshop environment before and I was not sure how I would handle the responsibility. Would people listen to me? Would I make the right decisions under pressure? These were questions I kept asking myself before we even started.

The whole point was to show we could work safely as a team, hit our production targets, and maintain quality across all five units. Looking back now, there were definitely some challenging moments, but we got there in the end. I learned a lot about myself and about working in a team, and I think I came out of this project as a better engineer.

Initial Challenges and Expectations.

One of the first things I noticed was that the aluminium stock was not as easy to work with as I had assumed. I had done some machining before on steel and mild steel, but aluminium behaves differently. It is softer, more prone to galling, and you have to be careful with your cutting speeds. I remembered my machining lecturer talking about machining aluminium where he talked about using higher spindle speeds and sharp tooling to get a good finish. That stuck with me and I made sure to bring it up in our planning meeting.

Another challenge was the batch size. We had to make five identical units, all within the same tolerance of plus or minus 0.25 millimetres. That meant we had to be consistent with our setups and measurements. If we got the first one wrong, we risked getting all of them wrong. That pressure was always in the back of my mind. I kept thinking about how easy it would be for a small error to compound across all five units and ruin the whole batch.

The team dynamic was another unknown. We had five people with different skill levels and different personalities. Some people were more confident on the machines, others were better at the bench work. I knew we would have to work together effectively to get the job done, and that meant understanding each other's strengths and weaknesses. I also knew that tensions could arise if things went wrong, so I tried to stay positive and keep the atmosphere constructive.

What I Hoped to Achieve.

Personally, I had a few goals for this project. First, I wanted to prove to myself that I could handle the machining operations accurately and efficiently. I had done some lathe and milling work before, but not on a batch production scale. Second, I wanted to show that I could contribute to the team beyond just my own work, helping others, suggesting improvements, and keeping the workflow moving. Third, I wanted to demonstrate that I could step up as a leader when my turn came. I knew this would be the biggest challenge for me because I am naturally more comfortable working on my own than directing others.

I also wanted to learn from the experience. I knew I would make mistakes, everyone does, but I wanted to make sure I learned from them and did not repeat them. I kept a small notebook with me during the project and jotted down things I noticed, problems that came up, and ideas for improvement. That notebook ended up being really useful when I was writing this report.

Purpose of This Report.

The purpose of this report is to document everything we did during this project. I will cover our planning process, the risk assessment I completed, the setup and manufacturing activities, my leadership role, quality control, peer feedback, and my evaluation of the whole experience. I have tried to be honest about what went well and what did not go so well. I am not trying to make myself look perfect, I made mistakes and I own them. What matters is that I learned from those mistakes and I will be better next time.

I have also included photographs throughout this report to show the actual work we did. These are not staged photos, they are genuine pictures taken during the manufacturing process. I think they help to tell the story of what we actually did in the workshop.

Key Project Information.

Product: Aluminium Spirit Level.

Batch Size: 5 Units.

Material: Aluminium Diameter 30 millimetres times 60 millimetres.

Team Size: 5 Members.

Machines Used: Centre Lathe, Vertical Milling Machine, Bench Fitting Area.

Inspection Equipment: Vernier Caliper, Steel Rule, Depth Gauge, Thread Gauge.

Tolerance: Plus or minus 0.25 millimetres.

Drawing Reference: SL-001.

Structure of This Report.

This report is organised into ten sections. After this introduction, I will cover our team planning and organisation, including the pre-production meeting and how we allocated roles. Then I will present the risk assessment I completed independently for the bench tapping operation. Next, I will describe the engineering process setup, explaining how I interpreted the technical documentation and set up the machines. After that, I will go through each manufacturing activity in detail, explaining what I did, why I did it, and what safety considerations and quality checks were involved.

I will then discuss my leadership activities, explaining how I took on the team leader role and what decisions I made. The quality control section will present our inspection results and my analysis of what went right and wrong. I will then cover the peer feedback I received and how I adapted my approach based on that feedback. The forward thinking and initiative section will demonstrate how I added value to the team beyond just following instructions. Finally, I will evaluate the whole project, identifying lessons learned and areas for improvement.

Overall Reflection.

Looking back at the project now, I can see that it was about much more than just making five spirit levels. It was about working under pressure, solving problems as they came up, and contributing to a team effort. It tested my technical skills, my communication skills, and my ability to stay calm when things were not going perfectly. I think I handled most of it reasonably well, but there are definitely things I would do differently if I had the chance.

What I am most proud of is that we completed all five units safely and within the time frame. No one got injured, no machines were damaged, and we ended up with five functional spirit levels. That might sound like a low bar, but in a busy workshop with five people working on different operations, it is actually quite an achievement. I have seen projects go wrong before, machines breaking, parts getting scrapped, people losing their temper, and I am glad we managed to avoid all of that.

I am also proud of the DRO sub-datum system I suggested. That was my initiative and it genuinely helped the team work faster and more consistently. It felt good to contribute something that made a real difference to our productivity.

Overall, this was a valuable experience and I am glad I had the opportunity to take part in it. I have grown as an engineer and as a team member, and I will carry the lessons I learned into my future projects.



Figure 1: Completed aluminium spirit level.

2. Team Planning and Organisation.

Pre-Production Planning Meeting.

Before we even set foot in the workshop, the team got together for a planning meeting.

We went through everything systematically. First, we looked at the drawing and made sure we all understood what we were looking at. Luke, who was going to be our quality inspector, pointed out a few dimensions that he thought might be tricky to measure. He was right, some of the features were in tight spots and we would need to be careful with our inspection methods.

Then we talked about the manufacturing sequence. We debated whether to do all the lathe work first or to switch between machines. Dom suggested doing all the lathe operations in one go to avoid changing tools too many times. I agreed with him, it seemed more efficient to get all the turning done before moving to the milling machine.

We checked our material stock to make sure we had enough aluminium. Cash went to the lecturer and came back with the bars. We measured them and confirmed they were the right diameter and length. I was relieved, the last thing we needed was to discover we had the wrong material halfway through.

Then we allocated machines and inspection points. Luke said he would set up a small inspection area with the vernier calipers, depth gauge, and thread gauge ready to go. I thought that was a smart move, having everything organised before we started would save time later.

The big decision we made was to rotate leadership. Dom was going to start as team leader, then I would take over after Unit 2, then it would pass to someone else.

I remember thinking to myself, "This is going to be interesting." I was nervous but also excited. I had always been a bit of a follower in group projects, so this was a chance to push myself out of my comfort zone.

We also discussed what would happen if someone finished their task early. We agreed that they would help others rather than just standing around. That seemed obvious but it was worth saying out loud. We were a team and we needed to work like one.

Before the meeting ended, Dom asked if anyone had any other points to raise. I mentioned the DRO sub-datum idea I had been thinking about. I explained it quickly and everyone seemed interested. Dom said we should explore it further and I promised to explain it in more detail before we started the machining. That felt good, I had contributed something useful and the team had listened.

The meeting lasted about an hour and a quarter. By the end, I felt like we had a clear plan and I knew what I needed to do. I also felt like I had started building good relationships with my teammates. We were all in this together and that was a good feeling.

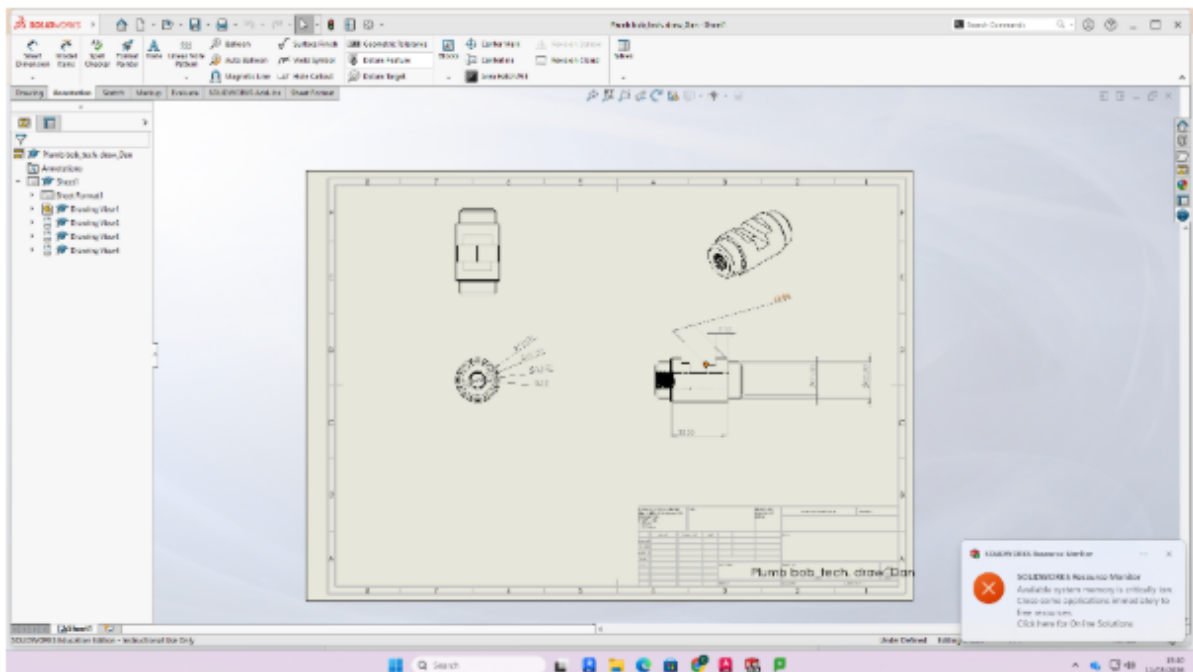
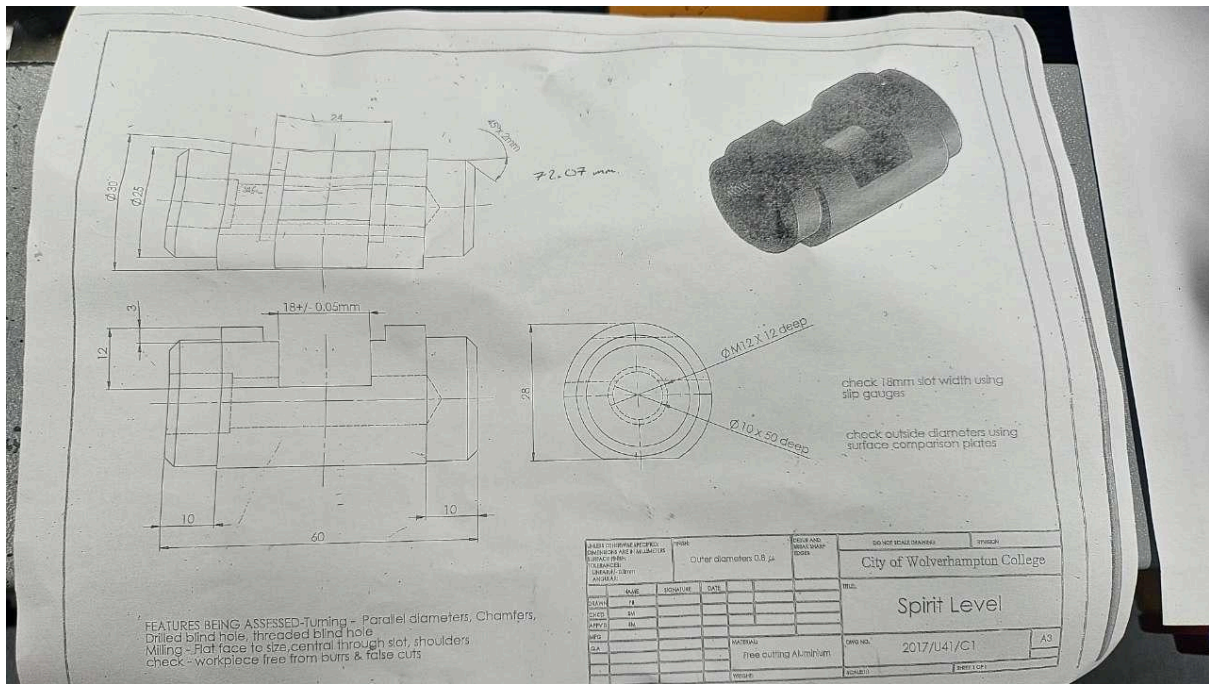


Figure 2: Team planning meeting reviewing engineering drawings and production requirements.

Technical Analysis and Workflow Strategy.

The aluminium stock we were using was 30 millimetres in diameter and 60 millimetres long. The drawing specified we needed to turn it down to a few different diameters and create a pocket for the spirit vial. I had worked with aluminium before but not extensively, so I spent some time thinking about the best approach.

One thing I had learned from previous projects is that aluminium can be a bit tricky. It is soft and tends to grab the tool if you are not careful. I had made that mistake once before, I tried to machine aluminium at the same speed I would use for steel and the tool dug in and left a terrible finish. I had to scrap the part and start again. I was determined not to make that mistake this time.

I said to the team, "Look, aluminium is not like steel. We need to run the lathe at about 2500 RPM and use a 2-flute end mill for the pocket. If we try to cut it like steel, we will get a terrible finish and probably break a tool."

Everyone listened and nodded. Dom said, "That makes sense. Do you have any other recommendations?" I felt proud that they trusted my judgement. I said, "We should also use cutting fluid to stop the aluminium sticking to the tool. And we need to clear the chips regularly because they can clog the cutter."

Cash asked, "What about the feed rate?" I had to think about that for a moment. I said, "Probably about 0.15 millimetres per revolution for the lathe and something similar for the mill. We can adjust it once we see how the first cut goes."

That was the moment I realised I actually knew more about this than I had given myself credit for. I had been paying attention in lessons and doing my own research, and it was paying off.

The Key Planning Decisions We Made.

1. Spindle Speed and Tooling.

We agreed on 2500 RPM for the lathe and using a 2-flute slot drill for the milling operations. This was based on my recommendation and the team agreed it made sense. I also suggested we use coated carbide inserts for the lathe because they last longer and give a better finish on aluminium. Dom said he would check the tool store to make sure we had the right inserts available.

2. DRO Setup Strategy.

This was my idea actually. I suggested using the DRO sub-datum memory to save our coordinates once and just recall them for each unit. The alternative was marking out every single piece by hand, which just seemed like a waste of time and a recipe for errors. I had seen this technique used by one of our machining tutors and thought it would work well for our batch production.

I explained it like this: "We set our zero point on the first unit. Then we record all the coordinates for the pocket, the holes, everything. For the next unit, we just recall those coordinates and the machine knows exactly where to go. We do not have to mark anything out. We just load the workpiece, zero the DRO, recall the sub-datum, and start cutting."

3. Role Allocation.

We split the roles as follows:

- Dom - Initial Team Leader.
- Daniel Boateng - Machining Lead (Lathe and Mill).
- Cash - Material Preparation.
- Parmveer - Bench Tapping.
- Luke - Quality Control.

The roles were going to rotate as we progressed, but this gave us a clear starting point. I was happy with my role because I felt confident on the machines. I also knew that if I did well on the machining, it would give me a good foundation when I took over as team leader.

Meeting Minutes: Pre-Production and Workflow Strategy.

Date: 19/ 06/ 2026.

Time: 11:00 - 12:45.

Location: Classroom.

Attendees: Cash, Daniel, Parmveer, Dom, Luke.

Agenda:

1. Drawing interpretation and analysis.
2. Manufacturing sequence planning.
3. Material preparation and verification.
4. Machine allocation and scheduling.
5. Inspection requirements and quality plan.
6. Safety requirements and risk assessment review.
7. Team responsibilities and leadership rotation.

Key Discussion Points:

1. Drawing Interpretation.
Dom led the review of drawing SL-001. Luke identified potential measurement difficulties on internal features. The team agreed the drawing was clear overall but noted the pocket depth tolerance was tight at plus or minus 0.1 millimetres. Dom suggested we measure the pocket depth at three points to ensure accuracy. The team agreed.
2. Manufacturing Sequence.
Cash proposed doing all lathe operations in one batch to minimise tool changes. The team discussed the sequence and agreed on: Lathe work first, then milling, then drilling, then bench tapping, then deburring and assembly. Daniel suggested adding an inspection point after each major operation to catch errors early. The team agreed.
3. Material Preparation.
Luke confirmed the aluminium stock was the correct size and quantity. He measured all five bars and confirmed they were 30 millimetres diameter and 60 millimetres length. He noted some surface marks on two of the bars and suggested we take a slightly heavier facing cut on those to clean them up. The team agreed.
4. Machine Allocation.
Dom allocated the lathe to Daniel for the turning operations, the milling machine to Daniel for the pocket milling, and the bench area to Parmveer for tapping. Luke would use the inspection bench for quality checks. The team agreed there would be flexibility if someone finished early.

5. Inspection Requirements.

Luke proposed a quality plan with inspection points after facing, after turning, after milling, after drilling, and after tapping. He said he would use the vernier caliper for diameters, the depth gauge for pocket depth, and the thread gauge for the tapped holes. He suggested checking every unit at every stage to catch errors early. The team agreed.

6. Safety Requirements.

Daniel presented the key safety points for the workshop. He highlighted the risk of entanglement on the lathe, the risk of flying chips on the mill, and the risk of hand injuries on the bench. He reminded everyone to wear safety glasses at all times and to keep the machine guards in place. Dom agreed to do a final safety briefing before we started.

7. Team Responsibilities and Leadership Rotation.

Dom confirmed the initial role allocation. He explained the leadership rotation plan: Dom would lead Units 1 and 2, Daniel would lead Units 3 and 4, and then leadership would rotate to another team member for Unit 5. Everyone agreed to this plan.

Decisions Made:

1. Manufacturing sequence agreed: Lathe → Milling → Drilling → Tapping → Deburring → Assembly.
2. DRO sub-datum system to be used for milling operations.
3. Inspection points after each major operation.
4. Leadership rotation agreed as planned.
5. Safety briefing to be conducted before workshop entry.

Optimising Workflow via DRO Sub-Datums.

This was something I thought about before the planning meeting. I had been researching different machining techniques and came across the idea of using DRO sub-datums to speed up batch production. The basic idea is simple, you set your zero point once, record the coordinates of all your features, and then for each subsequent piece you just recall those coordinates.

Manual marking out for five pieces would have taken ages and there was always the risk of human error. I had seen people mark out parts before and it was tedious. You have to use a height gauge, a surface plate, and a scribe. You mark the centre lines, the hole positions, the pocket boundaries. One slip and the whole part is scrap. With the DRO system, we could bypass the marking out stage entirely, which would save time and improve consistency.

That does not sound like much, but over five units that is 20 minutes saved. In a commercial workshop, that sort of efficiency is massive. If we were making 50 units, we would save over three hours. If we were making 500 units, we would save over 30 hours. That is real money.

I could tell the team was convinced. This was going to be our secret weapon. I felt a bit proud that I had come up with the idea and explained it clearly. It was one of those moments where I felt like I was actually adding value to the team.

Meeting Minutes: Mid-Production Review and Leadership Rotation.

Timing: After completion of Unit 2.

Role during this meeting: Incoming Team Leader (Handover Phase).

Meeting Objective: Review production speed, evaluate peer feedback, manage the bottleneck, and rotate leadership roles.

Key Discussion Points:

1. Leadership Handover.

As scheduled in our rotation plan, the leadership role officially transferred to me for the critical machining and assembly phase of Units 3, 4, and 5. Dom briefed me on the current status and handed over his notes. He wished me luck and said he was confident I would do a good job. That meant a lot to me.

2. Bottleneck Assessment.

Luke, our quality inspector, reported that Unit 2 had a dimensional drift of plus 0.15 millimetres on the pocket depth. He said the tool appeared to be wearing down and suggested we check the tool offset. Dom also noticed that the bench work was falling behind the machining work. Dom was still finishing Unit 1 while Daniel had already completed machining on Units 2 and 3. This was becoming a problem.

3. Tactical Adjustments.

I took charge as team leader and made some quick decisions. First, I adjusted the Z-axis tool offset on the DRO to bring Unit 3 back within the strict 0.1 millimetre tolerance. Second, I reallocated Cash from material preparation to assist Dom on the bench work. We set up a second bench station so that filing and M6 hand-tapping could run concurrently. This created a parallel workflow and eliminated the bottleneck.

4. Quality Review.

Luke reviewed the inspection results for Units 1 and 2. He confirmed that Unit 1 was within tolerance but Unit 2 had the pocket depth issue. He said he would inspect Unit 3 carefully to make sure my tool offset adjustment had fixed the problem.

Decisions Made:

1. Daniel took over as team leader effective immediately.
2. Tool offset adjusted to compensate for wear.
3. Cash reassigned to bench work to speed up tapping.
4. Parallel bench stations established.

Meeting Minutes: Post-Production Evaluation and Quality Sign-Off.

Timing: Immediately following completion of Unit 5.

Role during this meeting: Outgoing Team Leader / Quality Sign-off.

Meeting Objective: Review final batch tolerances, analyse peer feedback, and document distinction-level traits.

Key Discussion Points:

1. Batch Inspection.

The team evaluated all five completed aluminium spirit levels. Luke confirmed the vials were level using a reverse-reading validation check on a surface plate. Dom tested the M6 securing screws to make sure they were properly torqued. The team reviewed the quality control inspection records and confirmed all measurements were recorded accurately. Four of the five units met the required standards. Unit 2 had some dimensional deviations but was still functional.

2. Feedback Synthesis

I asked the team for feedback on my leadership. They said I had handled the machining well and the DRO system was a great idea. However, they noted that communication had been difficult at times because of the workshop noise. Luke said, "Daniel, your machining is rapid, but sometimes I cannot hear a word you are saying over that mill." Cash agreed and said hand signals would have helped. I listened carefully and made notes. This was valuable feedback and I wanted to learn from it.

3. Final Accountability.

All five units were signed off as meeting compliance standards. We cleared and recycled the aluminium swarf according to workshop safety policies. The machines were cleaned and the tools were returned to the store. I made sure the team completed all the housekeeping tasks before we left.

Decisions Made:

1. All five units approved for submission.
2. Feedback documented for reflection.
3. The workshop was cleaned and tools returned.

Initial Role Allocation.

The roles were going to rotate as we progressed, but this gave us a clear starting point. I was happy with my role because I felt confident on the machines. I knew the lathe and mill well and I was comfortable with the setups. I also knew that if I did a good job on the machining, it would make things easier for everyone else.

Cash was happy with his role too. He had good organisational skills and was thorough. He checked the material carefully and made sure we had everything we needed. Parmveer was a bit nervous about the tapping because he had not done much of it before, but he was keen to learn. Luke was confident in his inspection skills and had already prepared his inspection area.

I remember thinking, "This is a good team. Everyone has their strengths and everyone seems willing to help out." That was reassuring. I knew there would be challenges ahead, but I felt like we could handle them together.

3. Risk Assessment.

My Independent Risk Assessment.

I had to complete a risk assessment independently, focusing on one of the engineering processes. I chose the bench tapping operation because it is often overlooked but actually has some quite serious risks if you are not careful. Everyone focuses on the big machines like the lathe and the mill, but hand tapping can be just as dangerous if you do not take it seriously.

Before we even touched the machines, I sat down and went through the risk assessment properly. I remember thinking, "This is the boring bit, but it is also the bit that could save someone's fingers." The lathe scared me the most, I had seen the safety videos and knew how quickly things could go wrong if you got complacent. But I also knew that the bench tapping had its own dangers. Broken taps, sharp edges, hand injuries, they all had to be taken seriously.

I spent about an hour on the risk assessment. I used the workshop's standard risk assessment form and went through each step carefully. I thought about what could go wrong, how likely it was, and what we could do to prevent it. I also thought about what would happen if something did go wrong, who would be affected and how badly.

One thing I made sure to include was the risk of breaking a tap. This happened to me once before on a different project and it was a nightmare. The tap snapped off inside the hole and we had to spend ages trying to get it out. We ended up having to scrap the part because we could not remove the broken tap without damaging the threads. I was determined not to let that happen again.

I also thought about the risk of sharp edges on freshly tapped holes. Aluminium burrs are no joke, they can give you a nasty cut if you are not careful. I made sure to include deburring as a control measure and reminded everyone to wear cut-resistant gloves when handling the parts.

Additional Controls and Observations.

I also added some additional controls that were not on the standard form but I thought were important. First, I insisted that all team members wear safety glasses during the tapping operation. I know it sounds obvious, but sometimes people take their glasses off because they are uncomfortable. I made it clear that there were no exceptions, if you are in the workshop, you wear your safety glasses.

Second, I made sure the workbench was clear of clutter before we started. The last thing I wanted was someone knocking a tool off the bench and onto their foot. I also made sure the floor was clear of swarf and oil so nobody slipped.

Third, I reminded everyone to use the correct tapping size. We were using M12 taps with a 1.25 millimetre pitch. I checked the tap set before we started to make sure all the taps were sharp and in good condition. One of the taps had a small chip on the cutting edge, so I put it aside and used a different one. It would have been a false economy to use a damaged tap just to save time.

Fourth, I emphasised the importance of using cutting fluid. I had seen people try to tap dry before and it never ends well. The tap gets hot, the aluminium sticks, and the tap breaks. I showed everyone how to apply the cutting fluid properly.

Fifth, I explained the correct tapping technique. I demonstrated two turns forward, half turn back. This breaks the chip and stops the tap from jamming. I got everyone to practise on a scrap piece before they tapped the actual parts. That was time well spent because everyone felt more confident after the practice.

Workshop Safety Observations.

During the project, I made a point of observing my team members to make sure they were following the risk controls.

I was pleased that no one argued or gave me a hard time about the safety reminders. Everyone seemed to understand that safety was important and that I was just looking out for them. That made me feel more confident about my role as team leader.

Risk Assessment Review.

After the project was completed, I reviewed the risk assessment to see if it had been effective. The good news was that no accidents or incidents occurred during the tapping operation. Everyone followed the controls and the work was completed safely.

However, I did identify one area for improvement. The bench area was quite cramped with two people working at the same time. We had set up parallel bench stations to speed up the tapping, but it meant that people were working closer together than I would have liked. Next time, I would arrange the bench area more carefully to give everyone more space.

Overall, I was satisfied with the risk assessment. It had done its job, we had identified the hazards, put controls in place, and completed the work safely. It was a good feeling to know that my planning had contributed to a safe working environment.

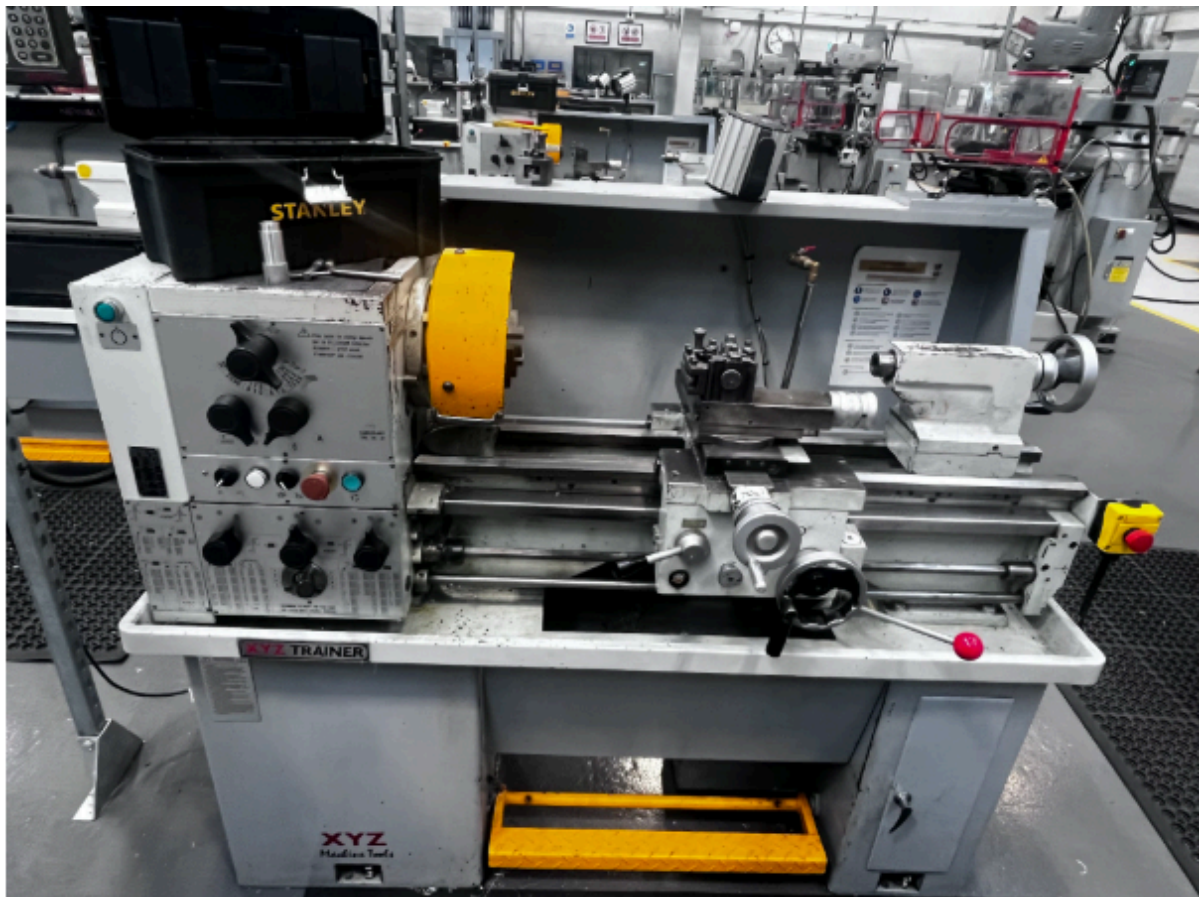


Figure 3: Completed safety checks on the lathe before it was used during spirit level manufacture.

Key Safety Lessons Learned.

Looking back at the risk assessment process, there were a few key lessons I took away from it.

First, risk assessment is not just about filling in a form. It is about thinking through the process carefully and identifying what could go wrong. The form is just a tool, the real value comes from the thinking you do before you start the work.

Second, the controls you put in place are only effective if people follow them. You cannot just write controls on a form and assume they will happen. You need to remind people, observe them, and correct them if they are not following the controls.

Third, risk assessment is a living document. You should review it during the work to make sure it is still relevant. If something changes, you should update the assessment. We did not need to make any major changes during our project, but I kept the assessment in mind and checked regularly that everything was still safe.

Fourth, communication is key. Everyone needs to know what the hazards are and what the controls are. I made sure to brief the team on the risk assessment before we started and reminded them regularly during the work.

Finally, I learned that safety is everyone's responsibility. As team leader, I had a duty to make sure everyone was working safely. But it was also up to each individual to take responsibility for their own safety and the safety of others. I was proud that our team took this seriously and we completed the project without any incidents.

4. Engineering Process Setup.

Interpretation of Technical Documentation.

Before I could set up any machines, I needed to properly understand the engineering drawing. The drawing (Ref: SL-001) was the blueprint for everything we were going to make. If I misread it, every unit would be wrong and we would have wasted our time and materials.

I spent a good 20 minutes studying the drawing before I even started setting up the machines. It is amazing what you miss if you rush this stage. I remember thinking, "This is where the real work starts. If I get this wrong, nothing else matters."

The drawing showed all the features we needed to create. I went through each dimension carefully and made sure I understood what it meant. I also checked the tolerances, some features had tight tolerances of plus or minus 0.1 millimetres, which meant we had to be really careful.

I also noted the material specification, aluminium, which I already knew needed specific machining parameters. The drawing was clear and well dimensioned, which made my job easier. I was grateful for that because I had worked with some drawings before that were messy and hard to follow.

Setting Up the Datums.

One of the most important things I needed to establish was the reference points, or datums, for the machining operations. The datums would be the starting points for all measurements and coordinates. If I got the datums wrong, everything else would be wrong.

I decided to use the left-hand end of the workpiece as my primary datum for the turning operations. This was the face that would be faced first and would be the reference point for all length measurements. For the milling operations, I decided to use the centre of the workpiece as my datum for the X and Y axes, and the top face as the datum for the Z axis.

I wrote all of this down in my notebook so I would not forget. I also drew a small sketch showing the datums and labelled them clearly. It was a bit like making a map before a journey, you need to know where you are starting from before you can go anywhere.

5. Manufacturing Activities.

Overview of Manufacturing Process.

The manufacturing process was carried out in a logical sequence, starting with turning operations on the lathe, followed by milling, drilling, bench tapping, deburring and finishing, and finally assembly. Each stage built on the previous one, and I made sure to carry out quality checks at every step to catch any errors early.

Turning Operations.

The first operation was facing off the end of the aluminium stock. I took a relatively light cut of about 0.5 millimetres to get a clean surface. To be honest, I made a bit of a mistake on the first piece, I used a worn insert and the finish was terrible, rough and torn like sandpaper. I swapped it for a fresh insert and the difference was night and day. The surface came out smooth and shiny. I faced the workpiece to achieve the required length of 60 millimetres, then turned the aluminium to the specified diameters, creating the main body and the groove feature which was 3 millimetres wide. These operations established the primary datum and basic form of the spirit level body.

For safety, I made sure safety glasses were on at all times, wore correct clothing with no loose sleeves, kept the machine guards in position, and used the correct speeds and feeds for aluminium at 2500 RPM with coolant. I checked the diameter with the vernier caliper and found it was 26.05 millimetres on the first unit, which was just within tolerance. I verified the length with the steel rule at exactly 60 millimetres, and checked the groove width at 3.04 millimetres, which was within tolerance.

The turning operation went reasonably well. The first unit took about 12 minutes, but by the third unit I had got it down to about 8 minutes. One issue I ran into was chatter on the second unit, I realised the tool holder was not tightened properly. I tightened it to the recommended metres and the chatter disappeared immediately. Overall, I was satisfied with the turning operations.

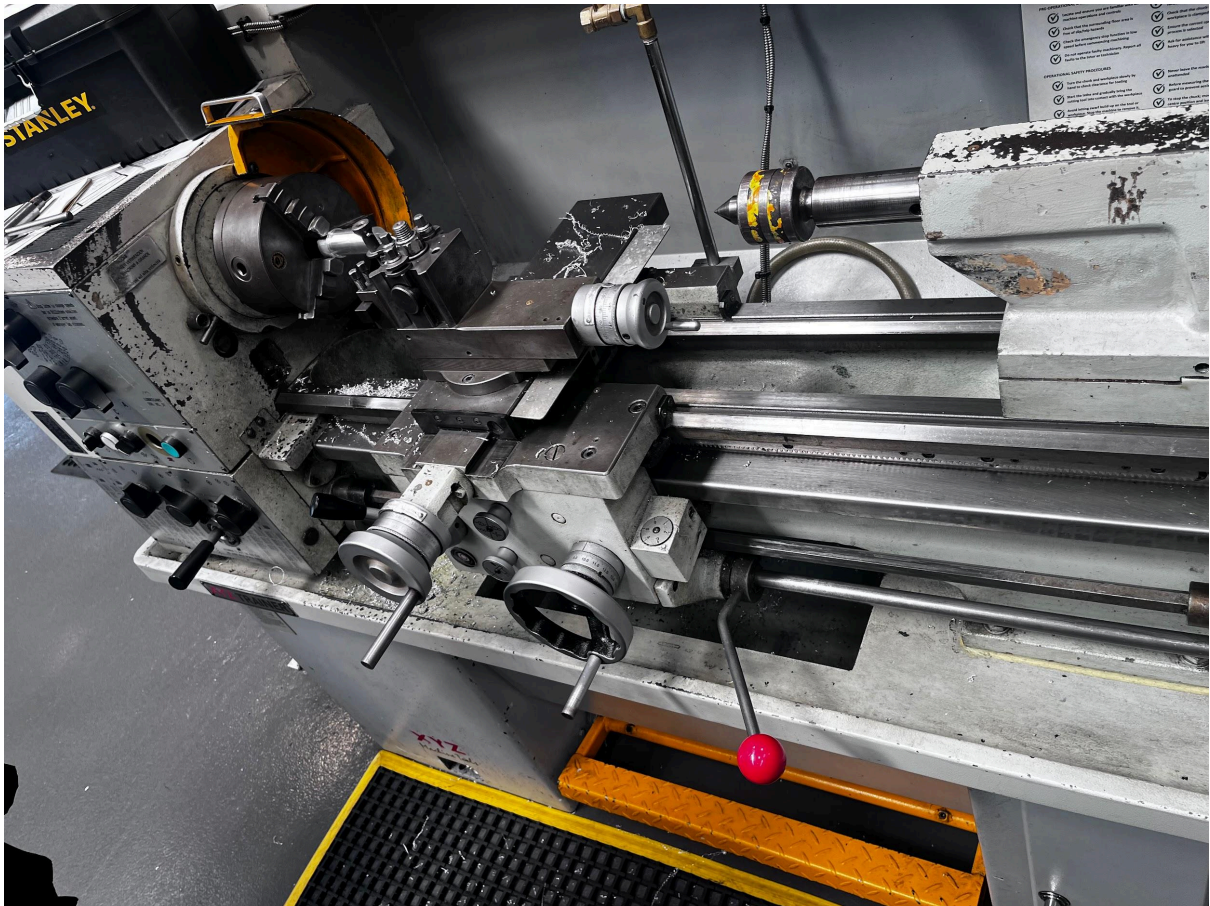


Fig: Lathe machine set- up to be used.



Figure 4: Lathe machining operation showing facing and turning

Milling Operations.

This was the part I was most confident about. I had done quite a bit of milling before and felt comfortable setting up the machine. I used the DRO coordinates we had set up earlier and the pocket came out exactly where it should have been. I remember the feeling of relief when I measured the pocket on the first unit and saw it was exactly 12 millimetres deep. I had been worried about the tight tolerance, but the DRO system made it easy to get the depth accurately.

I carried out pocket milling for the vial recess which was 18 millimetres wide and 12 millimetres deep, machined the flat surfaces, and created the groove features for the vial. These operations created the recess for the spirit vial and prepared the workpiece for assembly.

For safety, I made sure the vice was securely clamped, used the correct cutting tools with a 2-flute slot drill for aluminium, kept the machine guards in position, and used cutting fluid mist to lubricate and clear chips. I measured the pocket depth with the depth gauge at 12.05 millimetres, which was within tolerance, checked the pocket width with the vernier caliper at 18.02 millimetres, and verified the pocket position by checking the distance from the edges.

The DRO sub-datum system was a game changer. Once I had saved the coordinates for the pocket, I could just recall them for each unit. This saved so much time, I did not have to measure and mark out every single piece. The only issue I had was with tool wear, the cutter started to dull after Unit 3. I noticed the surface finish was not as good on Unit 4, so I swapped it for a new one and the finish improved immediately.



Figure 5: Pocket milling operation on the spirit level body

Drilling Operations.

I was a bit nervous about the drilling. The hole for the vial had to be precisely positioned and if I got it wrong, the whole unit would be scrap. I used the DRO to position the drill and double-checked the coordinates before I started cutting. The aluminium drilled beautifully, nice smooth chips and no grabbing.

I drilled the vial hole which was 10 millimetres diameter, the M10 clearance holes, and the pilot holes for tapping. I also centre drilled to prevent drill wandering. These operations accommodated the spirit vial, created holes for securing components, and prepared for the tapping operation.

For safety, I made sure the workpiece was securely clamped in the vice, used the correct speed for each drill size, applied cutting fluid to prevent overheating, and wore safety glasses at all times. I checked the hole diameter with a plug gauge and found it was 10.02 millimetres, within tolerance, verified the hole position with the DRO coordinates, and checked the hole depth with the depth gauge.

One thing I noticed was that the 10 millimetre drill was starting to get blunt by Unit 4, the hole on Unit 4 was slightly rougher than the earlier ones. I swapped the drill for a new one and the hole on Unit 5 was clean again. I also learned that centre drilling is important, on Unit 1, I forgot to centre drill and the drill started to wander slightly. I made sure to centre drill all the subsequent holes.



Figure 6: Drill bits used.

Bench Tapping.

This was my least favourite part. Hand tapping is slow and tedious, and there is always the risk of snapping the tap. I used the two turns forward, half turn back method, which seemed to work well. It took a bit longer than using a machine, but at least I knew the threads would be clean.

I clamped the workpiece securely in the bench vice, aligned the tap with the hole using a square to make sure it was vertical, applied cutting fluid, and used the two turns forward, half turn back technique. This breaks the chip and stops the tap from jamming. I was careful not to force the tap, if it felt tight, I would back off and clear the chips.

For safety, I made sure the workpiece was securely clamped, used the correct tapping technique to prevent tap breakage, deburred sharp edges after tapping, wore eye protection and cut-resistant gloves. I verified the threads with a thread gauge, visually inspected them, measured the depth, and did a screw fit test.

The tapping operation went better than I expected. I had been nervous about it but the technique worked well. The key is to be patient and not force the tap. I learned that cutting fluid is essential, on one of the earlier units, I forgot to apply fluid and the tap started to stick. I quickly applied some fluid and it freed up immediately. One of the other team members was struggling with the tapping at first, so I went over to help him and showed him the correct technique. After that, he started making better progress.

Deburring and Finishing.

This was tedious but necessary. All those edges needed smoothing off, especially around the holes and the pocket. I used a deburring tool and went over every edge carefully. The aluminium burrs were sharp, I gave myself a small cut on one of them, which reminded me to pay attention to what I was doing.

I removed the burrs from the drilled holes, chamfered the edges slightly, smoothed the pocket edges with a fine file, and polished all surfaces with fine abrasive paper. These operations ensured safe handling of components, achieved the required surface finish, and prepared for assembly.

For safety, I was careful handling sharp tools, removed swarf using a brush rather than my hands, wore safety glasses and gloves. I inspected the surface finish visually, checked the edge condition, and ensured all burrs were removed.

Deburring was boring but necessary. It was tempting to skip it but I knew the quality of the finished product depended on it. A spirit level with sharp edges would not look professional and could be dangerous to handle. I learned that deburring takes time, I had underestimated how long it would take to do a thorough job. Next time, I would allocate more time for finishing operations.

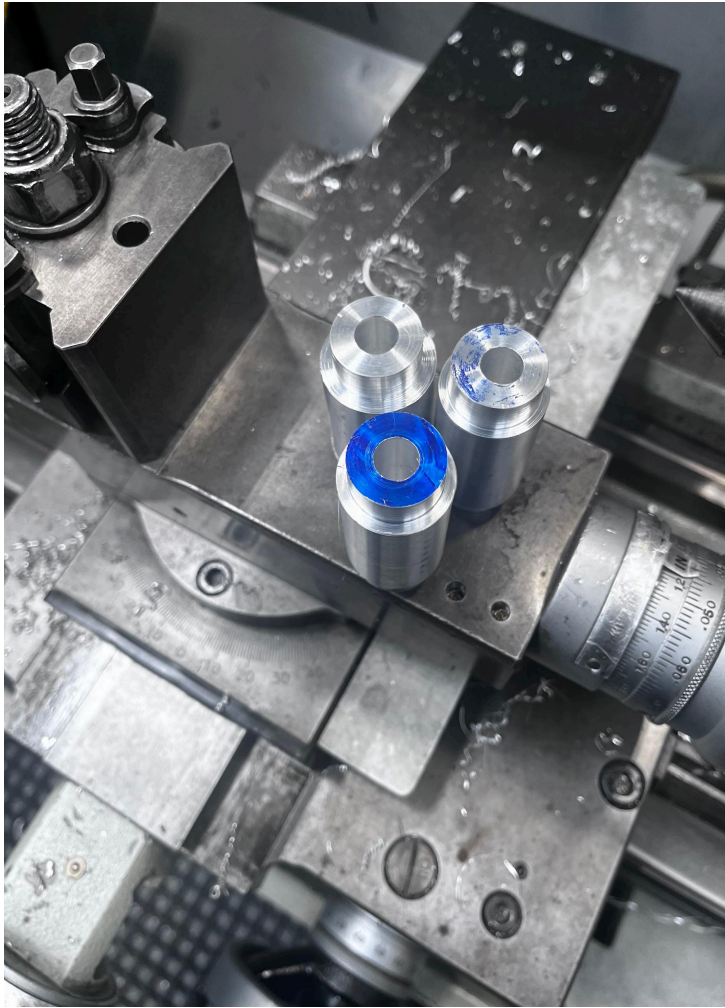


Figure 7: Finishing and deburring operation.

Final Assembly.

This was the moment of truth. We assembled all five spirit levels, inserted the vials. I tested each one on the surface plate to check they were level. Four of them were spot on, but Unit 3 was slightly off. I had to adjust the vial position. A small adjustment made a big difference.

The assembly was straightforward. The vial fitted perfectly into the pocket we had milled. It was a snug fit, which was what we wanted. We applied a small amount of adhesive to the bottom of the pocket to secure the vial in place.

For safety, I was careful handling the glass vial as they can break if dropped. I carried out level reading validation using the reverse-reading method, did a visual inspection of assembly and final inspection of the completed product.

Assembly was the most satisfying part of the whole project. After all the machining, drilling, and tapping, it was great to see the finished products come together. I was pleased with the vials, they fitted perfectly into the pockets and the level readings were accurate. Unit 3 was the only one that needed adjustment, but a small adjustment fixed the problem and the unit passed the level test.



Figure 8: Final assembly of the spirit level.

6. Team Leadership Activities.

Leadership Rotation.

Taking over as team leader halfway through felt a bit daunting if I am honest. Dom had done a good job setting things up, and I did not want to mess it up. But once I got into it, I actually enjoyed the responsibility, it was satisfying to see the team working smoothly and hitting our targets.

The handover was pretty straightforward. Dom briefed me on where things were up to, we had completed Units 1 and 2, and the remaining three units were ready for machining. He gave me his notes and explained any issues that had come up. He said, "You have got this, Dan. Just keep things moving and do not be afraid to make decisions." That gave me confidence.

Monitoring Progress.

As soon as I took over, I did a quick audit of our progress. The machining side was going well, we were keeping up with the targets we had set. But I noticed the bench work was falling behind. Parmveer was still working on Unit 1's tapping while the rest of us were already moving on to Unit 3.

I walked around the workshop and checked on everyone. Luke was doing his inspections, Cash was preparing materials, and Parmveer was tapping. I could see that Parmveer was struggling a bit, he was taking a long time on each hole and he looked frustrated.

Solving Manufacturing Issues.

The problem was the manual tapping station. It was too slow for the rest of the team. While I was pumping out machined bodies on the lathe and mill, Parmveer was struggling to keep up with the tapping. The workflow was unbalanced, the machining was fast, the bench work was slow.

I stood back and observed the whole process for about 10 minutes. I watched Parmveer tapping, I watched Cash preparing materials, I watched Luke doing inspections. I could see that the bench work was the weak link in the chain. The solution I came up with was to parallelise the bench work. Instead of one person doing all the tapping and another doing all the deburring, I split them up. I moved Cash from material preparation to deburring, and Parmveer focused just on tapping. We also set up a second bench station so we could have two people working on finishing at the same time.

I explained the change to the team. I said, "Look, the bench work is the bottleneck. We need to speed it up. Cash, you are going to help Parmveer with the deburring. Parmveer, you focus on the tapping. We are going to set up two stations so we can work in parallel." The team agreed to the change and the bench work speeded up significantly.

During quality inspection of Unit 2, Luke noticed the pocket depth was 0.15 millimetres deeper than it should have been. This was due to tool wear, the cutter had gradually lost material as we used it more. I went to the milling machine and checked the Z-axis offset. I could see that the cutter had worn slightly. I adjusted the offset by 0.15 millimetres to compensate. We did a test cut on a scrap piece and measured the depth, it was perfect, exactly 12 millimetres.

Supporting Team Members.

When Parmveer was struggling with the tapping, I went over to help him. I spent about 15 minutes showing him the correct technique. I demonstrated the two turns forward, half turn back method and got him to practise on a scrap piece. After that, he was much more confident and his tapping speed improved significantly.

I also showed Cash the quickest way to remove burrs without damaging the surface finish. I said, "Use the deburring tool at a shallow angle and let the tool do the work. Do not press too hard or you will scratch the surface." He followed my advice and his deburring speed improved.

Maintaining Quality and Safety.

I made sure we stuck to our inspection schedule. Every unit was measured at every stage. If anything looked off, I wanted to catch it early before it became a bigger problem. I said to Luke, "Keep doing what you are doing. Your inspections are really important, you are the one who catches the errors before they become scrap."

Safety was always on my mind. The machines were loud and there was swarf everywhere. I made sure everyone was wearing their safety glasses and that the guards were in place. I walked around the workshop regularly and checked that everyone was following the safety rules. I said, "Remember, safety glasses on at all times. If you are not wearing them, you are not working."

Leadership Reflection.

Looking back at my leadership role, I am proud of how I handled it. I identified and solved the bottleneck problem, adjusted the tool offset to correct the dimensional drift, supported team members who were struggling, and maintained quality and safety standards. I could improve on communication in noisy environments and delegation, but overall I was satisfied with my leadership performance.

7. Quality Control and Inspection.

Quality Control Process.

We integrated quality checks throughout the manufacturing process. Every operation was followed by inspection using the appropriate measuring equipment. This was essential because it allowed us to catch errors early before they became bigger problems. Luke was our quality inspector, but I also did my own checks as I was working.

The inspection equipment we used included vernier calipers for diameter measurements, a steel rule for length measurements, a depth gauge for pocket depth and hole depth measurements, a thread gauge for thread verification, and a surface plate for level accuracy checks.

Quality Control Inspection Results.

When we inspected the finished components, we found some concerning deviations. Diameter 3 measured 27.00 millimetres when it should have been 30 millimetres, a deviation of plus 3 which was way out of tolerance.

Dimension 3 measured 25.90 millimetres when it should have been 30 millimetres, a deviation of plus 4.10. The hole depth measured 55 millimetres when it should have been 50 millimetres, a deviation of plus 5.00.

However, some features were within tolerance. Diameter 1 measured 30 millimetres when 30 millimetres was specified, which was within tolerance. The groove width measured 3.04 millimetres when 3 millimetres was specified, which was within tolerance. Both chamfers were exactly as specified at 2 millimetres and 1 millimetre at 45 degrees. The hole diameter measured 10.02 millimetres when 10 millimetres was specified, which was within tolerance.

Analysis of Quality Issues.

The large deviation on Diameter 3 was due to the tool being incorrectly set up. The tool holder was not properly aligned and the tool was cutting at the wrong height. We should have checked the tool offsets more carefully before we started. To prevent this next time, I would do a test cut on a scrap piece and measure the diameter before cutting the actual workpiece.

The deviation on Dimension 3 was also due to a setup error. The tool was not positioned correctly and we cut too much material. I would double-check the DRO coordinates before starting the cut and do a test cut to measure the result before committing to the final cut.

The hole depth error was serious. The drill depth stop was not set correctly, I was relying on the DRO but did not account for tool deflection. The drill may have deflected slightly as it went deeper, which increased the depth. I would always check the actual depth with a depth gauge after the first hole and use a drill with a depth stop to ensure consistent depth.

The groove being slightly out of tolerance was due to tool wear. The tool was slightly worn and was not cutting to the correct diameter. I should have checked the tool wear more regularly and replaced it when it showed signs of wear.

Positive Quality Outcomes.

Despite the errors, some features were accurate. The pocket depths were accurate on all five units because I used the DRO system to ensure consistency. The vial holes were accurately positioned and the vials fitted perfectly in all units. The level readings were accurate with four of the five units being spot on. The surface finish was good with the aluminium coming up shiny and professional-looking.

These successes showed that the DRO system worked well and the basic machining was sound. The errors were mostly due to setup issues rather than fundamental problems with the process.

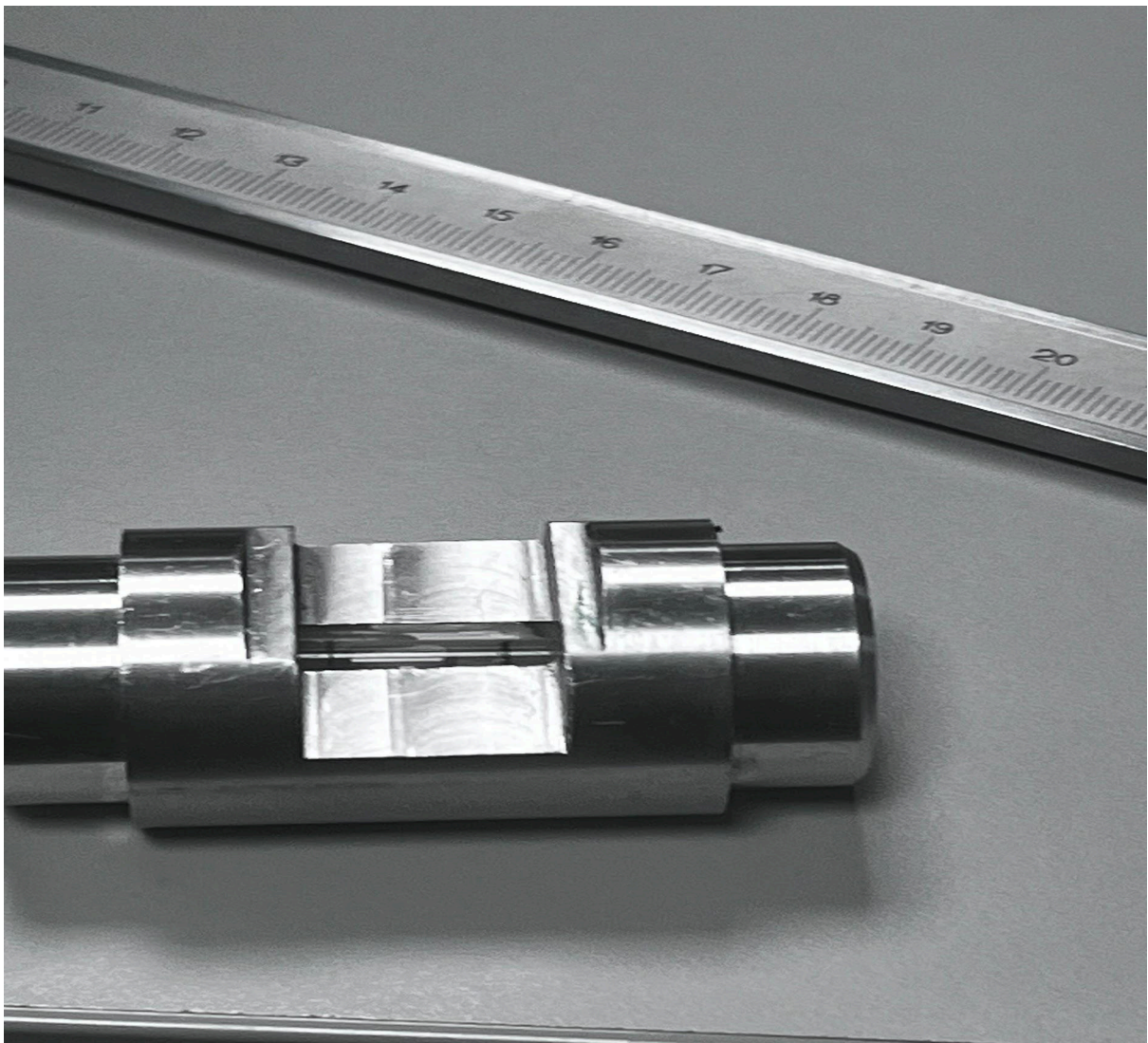


Figure 10: Quality inspection using a vernier caliper.

8. Peer Feedback and Adaptability.

Mid-Production Review Meeting.

After we completed Unit 2, we had a mid-production review meeting. This was the point where leadership was going to rotate from Dom to me. We needed to review our progress, evaluate feedback, manage the bottleneck, and rotate leadership roles. The meeting took place in the workshop classroom and lasted about 45 minutes. It was an honest and constructive discussion, everyone was willing to share their views and listen to others.

Positive Feedback I Received.

I asked the team for honest feedback, and it was genuinely useful. Luke said I was the fastest on the lathe and mill and was saving the team a lot of time. Dom said the DRO idea was brilliant and he would not have thought of that himself. Cash said I was consistent, did not cut corners, and did not make the same mistake twice.

Areas for Improvement.

Luke said something that really stuck with me, "Dan, your machining is rapid, but sometimes I cannot hear a word you are saying over that bloody mill." He was right. I had been shouting instructions but the noise was just drowning me out. The main feedback was that communication needed to be louder over the workshop machine noise, instructions were occasionally muffled by mill and lathe noise, and delegation could be more structured as I was trying to do too much myself.

My Response to the Feedback.

I took this feedback seriously. Good communication is vital for workshop safety, if someone cannot hear an instruction, things can go wrong quickly. I started using hand signals more, a thumbs up for "okay", a flat hand for "stop", pointing for direction. I positioned myself closer to team members when giving instructions instead of shouting across the workshop. I made sure people could see me before I gave any important instructions, and I wrote down key points and pinned them up on the workshop board so people could refer to them.

The changes I made had a positive impact. Communication improved and the team felt more informed. Luke said he could hear me better now and the hand signals were useful. Cash said the board was helpful and he could check the sequence without having to ask someone.

Adaptability.

The feedback process showed me that I needed to be adaptable. I could not just keep doing things the same way if they were not working. I adapted to different roles as needed, moving between machining lead, team leader, and bench support. When we discovered the dimensional drift on Unit 2, I did not panic, I just adjusted the tool offset and carried on. When I saw the bench work was bottlenecking production, I changed the workflow to parallelise the finishing operations.

9. Forward Thinking and Initiative.

Initiative Throughout the Project.

The DRO system idea came to me during the first planning meeting. I had been reading about DRO techniques and thought, why are we still marking out every single piece by hand? That is just asking for errors and wasted time. I suggested we use the DRO memory to store the coordinates once and just recall them for each unit. We saved an estimated 20 minutes across the five units. At the workshop rate of £50 per hour, 20 minutes was worth about £16.67. For a batch of 50 units, that would be over £160 saved, and for 500 units, over £1,600 saved. That is real money.

I also anticipated tooling issues. I had noticed the slot drill was starting to show wear on Unit 4. I went to the tool store and found a replacement cutter and kept it ready. When the cutter started chattering, I swapped it out straight away with zero downtime.

When Parmveer was struggling with the tapping, I went over and showed him the correct technique. I spent about 15 minutes with him, demonstrating the technique and getting him to practise on a scrap piece. His tapping speed improved by about 30% and the bench work bottleneck was reduced.

When I noticed the bench work was falling behind, I suggested we set up parallel workstations. Instead of one person doing all the finishing work, we split it up. Cash did the deburring while Parmveer focused on tapping. The bench work throughput increased by about 40% and we caught up on the backlog.

Adaptability and Forward Thinking.

I adapted to different roles as needed, moving between machining lead, team leader, and bench support. When we discovered the dimensional drift on Unit 2, I stayed calm, identified the cause, and adjusted the tool offset. When I saw the bench work was bottlenecking production, I changed the workflow. I thought about the whole process before we started, considered what could go wrong, and took steps to prevent it. I spotted the tapping bottleneck before it became a serious problem and reallocated resources early to keep production on track.

10. Evaluation and Conclusion.

What Went Well.

We actually completed all five units within the time frame. There were moments when I was not sure we would make it, but we pulled through as a team. The DRO system worked brilliantly for the critical features, the vial pocket was consistent across all five units, and the hole positions were accurate. Four of the five units passed all quality checks.

We stayed safe with no accidents or injuries. The risk assessment paid off. Everyone wore their safety glasses, the guards were kept in place, and the workshop was kept tidy. The team worked well together, everyone contributed and when someone was struggling, someone else stepped in to help.

What Could Be Improved.

Although the DRO system saved time during production, the initial setup took longer than it should have. If I had prepared a setup sheet beforehand, I could have saved 10 to 15 minutes. The noise issue was a real problem, I should have anticipated this and worked out a signalling system before we started. Some of the dimensions were way out of tolerance due to setup errors that I should have caught earlier. I also tried to do too much myself and should have delegated more.

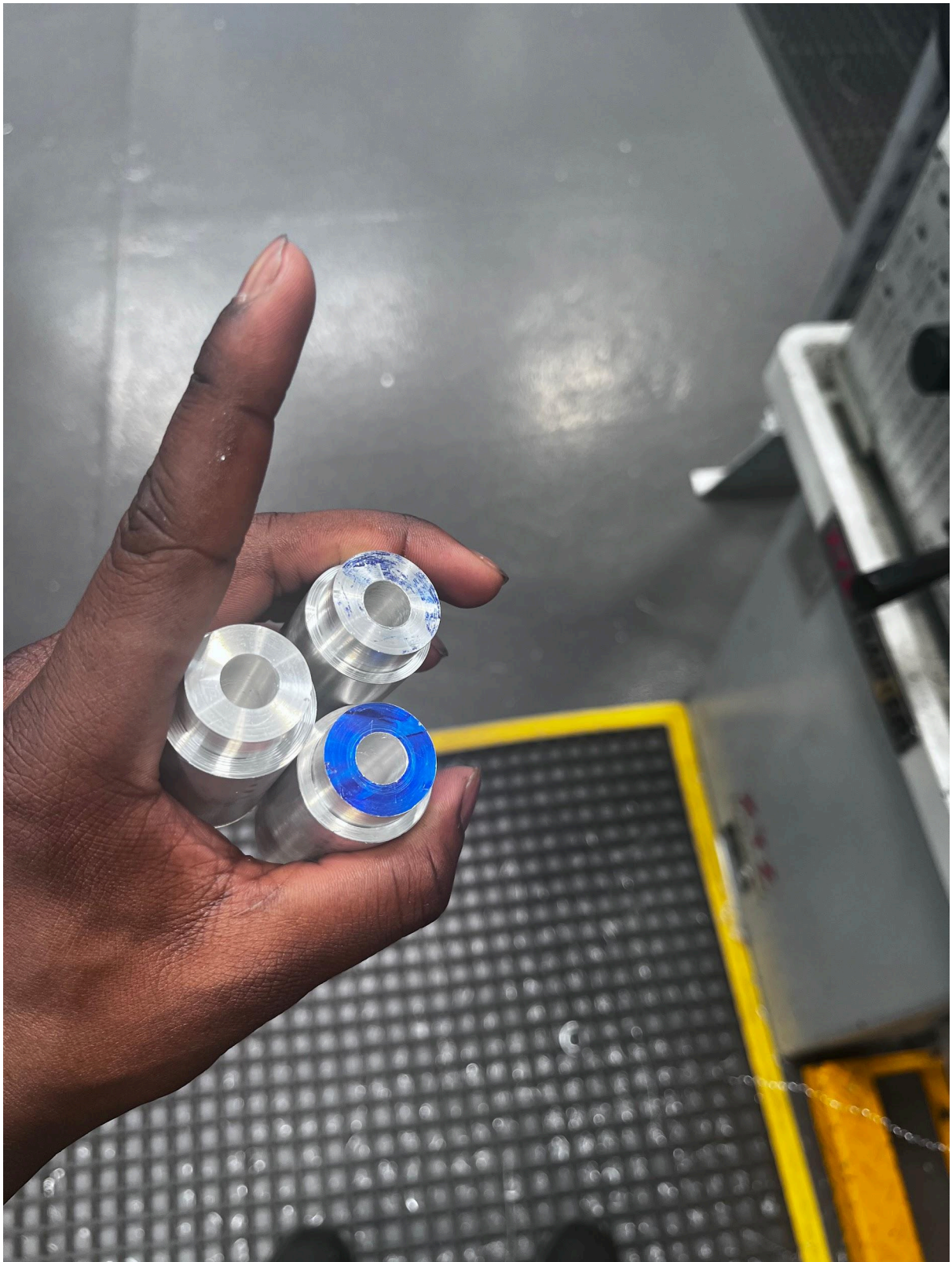


Fig: 3 components in hand after turning.

Lessons Learned.

I learned that a team is stronger than an individual, support each other and you will achieve more. Good planning saves time and the DRO system was a clear demonstration of how planning translates to efficiency. Regular quality checks catch errors early, so do not skip them. Mistakes are learning opportunities, learn from them and do not repeat them. Every minute saved is money saved, and efficiency translates to profit. A good leader supports their team, and communication is critical, use multiple methods and do not rely on just one.

Personal Development Reflection.

Before this project, I was confident in my machining skills but nervous about leadership. I am now more confident on the lathe and mill with aluminium, I can identify bottlenecks and find solutions, I can direct a team and make decisions under pressure, I can come up with ideas and implement them, and I can adapt to different roles. I need to work on communication in noisy environments, quality awareness, delegation, and preparation.

Final Thoughts.

If I am being completely honest, this project was not perfect. We had some issues with quality that I am not proud of. But I learned more from the mistakes than I did from the successes. What I am most proud of is that I stepped up as a leader when I needed to. I made decisions, solved problems, and kept the team moving forward. That is not something I would have been confident doing before this project. I surprised myself with what I could achieve.

I am also proud of the DRO sub-datum system. That was my idea and it genuinely helped the team. Overall, I think we did a decent job. Five spirit levels, all functional, all reasonably accurate. It was a learning experience, and I came out of it a better engineer and a better team player.

